

AMENDMENT NO. 1 JUNE 2024

TO

IS 14220 : 2018 OPENWELL SUBMERSIBLE PUMPSETS — SPECIFICATION

(First Revision)

(Second cover page, foreword) — Insert the following before the last paragraph:

‘Graphs for calculating minimum pump efficiency are replaced by two separate equations. Equation (1) for horizontal single stage volute type openwell submersible pumpset is adopted from Commission Regulation (EU) No. 547/2012 with regarding to eco-design requirements for water pumps for minimum efficiency index (MEI), for pump type end suction close coupled (ESCC). The equation (2) is for vertical/horizontal three or more stages diffuser type open well submersible pumpsets’

(Page 4, clause 9) — Substitute the following for the existing:

‘9 EFFICIENCY

The pump efficiency of open well submersible pumps at duty point as declared by the manufacturer for 4 and 2 poles shall not be less than calculated as in **16.4.1.**’

(Page 8, clause **16.1.2**, line 4) — Substitute ‘However, after applying the tolerance, overall efficiency value shall not be less than that obtained after multiplying the motor efficiency factor and pump efficiency as derived from equation (1) of **16.4.1.**’ for ‘However, after applying the tolerance, overall efficiency value shall not be less than that calculated from Fig. 5, Fig. 6 and Fig. 7.’

[Page 9, clause **16.4.1**] — Substitute the following for the existing:

‘**16.4.1** Pump efficiency shall be calculated using the equation as given below, and efficiency corresponding to MEL 0.2 shall be taken as the minimum pump efficiency:

- a) Pump efficiency, Horizontal single stage volute type openwell submersible pumpset (Fig. 2 typical).

$$\eta_p = 88.59 x + 13.46 y - 11.48 x^2 - 0.85 y^2 - 0.38 xy - C \dots(1)$$

where

η_p = minimum pump efficiency;

$x = \ln(n_s)$;

$y = \ln(Q)$;

$\ln$ = natural logarithm;

n_s = specific speed; and

Q = ‘discharge rate’ in m³/h.

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

where

n = ‘rotational speed’ in revolutions per minute (rpm);

H = ‘head’ per stage in meters (m);

Q = ‘discharge rate’ in m³/h; and

C = values for different pump efficiency levels.

Price Group 2

Amendment No.1 to IS 14220 : 2018

Minimum Efficiency Level, MEL		0.20	0.30	0.40	0.50	0.60	0.70
C	2 Pole, Speed	133.82	132.23	130.77	129.86	128.8	127.75
	4 Pole, Speed	131.2	129.77	128.46	127.38	126.57	125.46

NOTES

- 1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.
- 2 An illustrative example for calculating pump efficiency using above equation is given in Annex B.
- 3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.

- b) Pump efficiency, vertical/horizontal three or more stages diffuser type multistage open well submersible pumpsets, 2 pole and 4 pole speed (Fig. 3 typical).

$$\eta_p = 0.6571x + 0.0851y - 0.00534x^2 - 0.000565y^2 + 0.00106xy + C \dots(2)$$

where

- η_p = minimum pump efficiency;
 $x = n_s$;
 $y = Q$;
 n_s = specific speed;
 Q = 'discharge rate', in m³/h; and
 C = values for different pump efficiency levels.

Minimum Efficiency Level, MEL	0.20	0.30	0.40	0.50	0.60	0.70
C	42.0	44.0	45.9	47.7	49.4	51.0

NOTES

- 1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.
- 2 An illustrative example for calculating pump efficiency using above equation is given in Annex C.
- 3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

where

- n = 'rotational speed' in revolutions per minute (rpm);
 H = 'head' per stage in meters (m); and
 Q = 'discharge rate' in m³/h.

(Page 9, clause 16.4.1) — Insert the following new clause at the end:

‘16.4.2 The efficiency calculated as per 16.4.1(b) represents three or more stages of vertical and horizontal multistage diffuser type open well submersible pumpsets, 2 pole speed.

- a) For two stage diffuser type pump, multiply efficiency calculated by a factor 0.98; and
- b) For single stage diffuser type pump, multiply efficiency calculated by a factor 0.97.’

(Page 9, clause 16.5.1) — Substitute the following for the existing:

‘16.5.1 For arriving overall efficiency percentage (minimum) of the pumpset, multiply pump efficiency (minimum) value calculated as per 16.4.1(a) or 16.4.1(b) by motor efficiency factor.’

Amendment No.1 to IS 14220 : 2018

(Page 14, Fig. 5) — Delete.

(Page 15, Fig. 6) — Delete.

(Page 16, Fig. 7) — Delete.

(Page 11, Annex A) — Insert the following at the end:

‘ANNEX B

(Clause 16.4.1)

**EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR HORIZONTAL SINGLE
STAGE END SUCTION VOLUTE TYPE OPENWELL SUBMERSIBLE PUMPSET (2 POLE) DUTY
POINT**

B-1 INPUT DATA

Head = 32 m
Discharge rate = 6.5 lps
Speed = 2900 rpm

B-2 CALCULATIONS

B-2.1 To Find out Discharge Rate, Q, in m³/h

Discharge rate = 6.5 lps × 3.6 = 23.4 m³/h.

B-2.2 To Find out Specific Speed, n_s

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 2900 \times \frac{\sqrt{23.4/3600}}{(32)^{0.75}} = 17.37$$

NOTE — For specific speed calculation we have to use discharge (Q), in m³/h and head per stage (H) in m.

B-2.3 To Find out Minimum Pump Efficiency, η_p

$$\eta_p = 88.59x + 13.46y - 11.48x^2 - 0.85y^2 - 0.38xy - C \quad \dots (1)$$

$$x = \ln(n_s) = \ln(17.37) = 2.85$$

$$y = \ln(Q), = \ln(23.4) = 3.15$$

NOTE — For minimum pump efficiency calculation we have to use discharge (Q), in m³/h.

C = 133.82 (C-value for MEL 0.2 to be taken from the table for minimum efficiency calculation for 2 pole)

$$\eta_p = 88.59 \times 2.85 + 13.46 \times 3.15 - 11.48 \times 2.85^2 - 0.85 \times 3.15^2 - 0.38 \times 2.85 \times 3.15 - 133.82$$

$$\eta_p = 55.97 \% \text{ (Minimum pump efficiency)}$$

Amendment No.1 to IS 14220 : 2018

ANNEX C

(Clause 16.4.2)

EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR VERTICAL AND HORIZONTAL MULTISTAGE DIFFUSER TYPE OPENWELL SUBMERSIBLE PUMPSET (2 POLE) DUTY POINT

C-1 INPUT DATA

Head = 32 m
Discharge = 6.5 lps
Speed = 2900 rpm
Stages = 02

C-2 CALCULATIONS

Head per stage = $32/2 = 16$ m.

C-2.1 To find out Discharge Rate, Q, in m³/h

Discharge = $6.5 \text{ lps} \times 3.6 = 23.4 \text{ m}^3/\text{h}$

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 2900 \times \frac{\sqrt{23.4/3600}}{(16)^{0.75}} = 29.22$$

NOTE — For specific speed calculation we have to use discharge (Q), in m³/h and head per stage (H) in m.

$$\eta_p = 0.6571x + 0.0851y - 0.00534x^2 - 0.000565y^2 + 0.00106xy + C \quad \dots(2)$$

$$x = n_s = 29.22$$

$$y = Q = 23.40$$

NOTE — For minimum pump efficiency calculation we have to use discharge(Q) in m³/h.

C = 42 (C-value for MEL 0.2 to be taken from the table for minimum pump efficiency calculation)

$$\eta_p = 0.6571 \times 29.22 + 0.0851 \times 23.4 - 0.00534 \times 29.22^2 - 0.000565 \times 23.4^2 + 0.00106 \times 29.22 \times 23.4 + 42$$

$$\eta_p = 59.05 \% \times 0.98 \text{ (Factor for 2 stage)}$$

$$\eta_p = 57.86 \% \text{ (Minimum pump efficiency)}$$

(MED 20)